



IILM UNIVERSITY, GREATER NOIDA

INTERNSHIP REPORT PRESENTATION

NetSage AI

AI Network Troubleshooting Assistant

Cisco Virtual Internship Program | Artificial Intelligence

Abhay Shakya

Roll No. 2410031084 | B.Tech CSE

May – August 2026



**Abhay
Shakya**

B.Tech – Computer Science &
Engineering

ABOUT THE STUDENT

Candidate Profile



Roll Number

2410031084



Institute

IILM University, Greater Noida, U.P.



Programme

B.Tech CSE, Batch 2023–2027



Internship Domain

Artificial Intelligence



Duration

May – August 2026

Cisco Virtual Internship Program

The internship provided a practical environment to apply Artificial Intelligence and software-development concepts to a network troubleshooting problem. The project focused on building NetSage AI, an assistant that combines AI-generated diagnosis with deterministic validation, human review and evidence-based verification for Cisco-style Packet Tracer scenarios.

AI

Artificial Intelligence

Apply AI models to symptoms, topology notes and command evidence for structured diagnosis.

NW

Network Troubleshooting

Use Cisco Packet Tracer scenarios and show-command outputs to identify likely faults.

✓

Human-in-the-Loop

Keep diagnosis reviewable: accept, edit or reject before any recommendation is accepted.

Turning Network Symptoms into Trustworthy Diagnosis

Network troubleshooting requires learners to interpret symptoms, inspect topology and command outputs, choose the next diagnostic step and connect evidence to the underlying fault. A generative AI response can sound convincing while still being wrong. NetSage addresses this gap by combining AI diagnosis with deterministic checks, human review and explicit verification.



What NetSage Was Built to Achieve

AI

AI Diagnosis

Generate a structured diagnosis from symptoms, topology and evidence.

>_

Next Command

Recommend the most useful Cisco diagnostic command for the case.

R

Rule Validation

Cross-check AI findings against deterministic network rules.

H

Human Review

Require Accept / Edit / Reject before accepting the diagnosis.

S

Safe Remediation

Keep proposed fixes advisory; never auto-apply network changes.

V

Verification

Use post-fix evidence before marking a troubleshooting case verified.

The Toolkit Behind Net Sage AI



Python

Core implementation



LLM / AI Engine

Structured diagnosis



Ollama / OpenAI

AI inference options



Streamlit

Interactive review UI



Cisco Packet Tracer

Network simulation



CSV / JSON

Cases, results & audit data



Pytest

Automated testing



HTML / CSS

Dashboard presentation

From Evidence to Verified Diagnosis



The NetSage Development Journey

1

Define the Problem

Map common Packet Tracer faults and required outputs.

2

Design the Schema

Structure symptom, evidence, diagnosis and review fields.

3

Build AI Diagnosis

Generate evidence-grounded, structured troubleshooting output.

4

Add Rule Checker

Validate confidence, commands, evidence and safe fixes.

5

Build Review Gate

Force a human decision before acceptance.

6

Add Verification

Record post-fix evidence separately from diagnosis.

7

Run Test Cases

Evaluate the workflow across 44 reference-based cases.

8

Generate Dashboard

Summarize outcomes and maintain an audit trail.

Evidence-First Evaluation

44

Reference-based cases

23

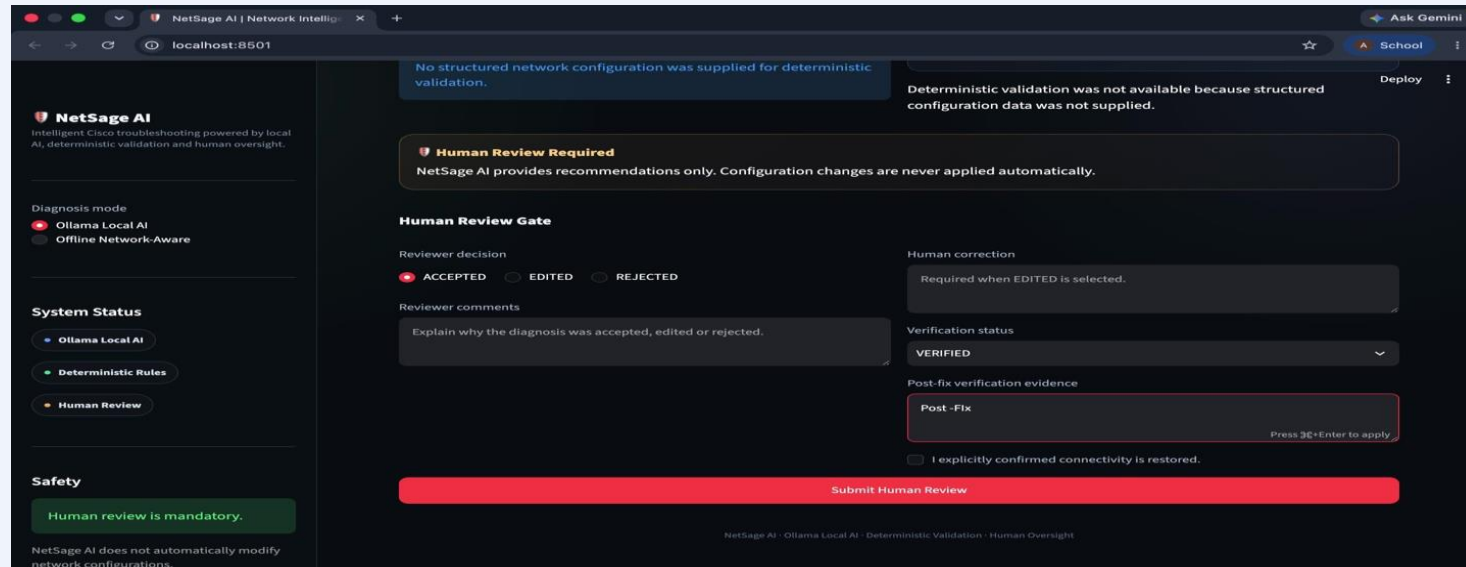
Layer 2 cases

20

Layer 3 cases

1

Layer 1 case



Mandatory Human Review

Every diagnosis passes through a review gate with three outcomes: Accepted, Edited or Rejected. Human correction is recorded separately from the AI diagnosis.

44 accepted after review

Acceptance is a reference-case evaluation result, not a claim of live-network AI accuracy.

EXPECTED OUTCOMES

Skills & Capabilities Demonstrated

- ✓ **AI-assisted troubleshooting** Convert network symptoms and evidence into structured diagnostic recommendations.
- ✓ **Explainable reasoning** Show likely fault, OSI layer, confidence and evidence-backed next action.
- ✓ **Responsible AI workflow** Separate AI generation from validation, human acceptance and verification.
- ✓ **Network domain knowledge** Apply concepts including VLAN, trunking, STP, ACL, NAT and routing.
- ✓ **Software engineering** Build modular Python components, tests, data pipelines and reporting.
- ✓ **Interactive dashboard** Present case-level decisions and aggregate results through a visual interface.

INTERNSHIP PROJECT

Thank You

NetSage AI

AI Network Troubleshooting Assistant
Human-in-the-Loop Diagnosis • Evidence • Verification

Abhay Shakya | Roll No. 2410031084

B.Tech CSE | IILM University